Triple quantum K-theory of flag varieties

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October 24, 2026

Abstract

We prove a triple quantum K-Pieri formula expanding the product of a quantum double Grothendieck polynomial and a quantized top block. Its coefficients are indexed by endpoints of admissible chains in the quantum Bruhat graph.

1 Polynomial conventions

Let S_∞ denote the permutations of the positive integers with finite support. We use one-line notation, let $\ell(w)$ be the inversion number of w, and write t_{ab} for the transposition interchanging positions $a < b$. Thus wt_{ab} is obtained from w by interchanging $w(a)$ and $w(b)$.

Let β, z, and the entries of $x = (x_1, x_2, \ldots)$, $y = (y_1, y_2, \ldots)$, and $q = (q_1, q_2, \ldots)$ be independent indeterminates. Put

$$X_i = 1 + \beta x_i, \qquad Q_i = \beta^2 q_i, \qquad \overline{y}_a = -\frac{y_a}{1+\beta y_a}.$$

Our formal group law is $a \oplus b = a + b + \beta ab$, so that $a \oplus \overline{a} = 0$. In particular, $z + \overline{y}_a$ below denotes an *ordinary* sum, not $z \oplus \overline{y}_a$.

We normalize the double Grothendieck polynomials by

$$\mathfrak{G}_{w_0^{(n)}}(x;y) = \prod_{\substack{i,j\geq 1\\ i+j\leq n}} (x_i \oplus y_j),$$

and the isobaric divided-difference recursion

$$\mathfrak{G}_{ws_i}(x;y) = \pi_i^\beta \mathfrak{G}_w(x;y) \quad \text{if } \ell(ws_i) < \ell(w), \qquad \pi_i^\beta f = \frac{(1+\beta x_{i+1})f - (1+\beta x_i)s_i f}{x_i - x_{i+1}}.$$

Here s_i interchanges x_i, x_{i+1} and fixes all other variables. In this convention $\mathfrak{G}_{21} = x_1 + y_1 + \beta x_1 y_1$, and $\mathfrak{G}_w|_{\beta=0}$ is the double Schubert polynomial with second alphabet $-y$.

For $0 \leq r \leq j$ define

$$F_r^j(X;q,\beta) = \sum_{\substack{J\subseteq\{1,\ldots,j\}\\ |J|=r}} \left(\prod_{a\in J} X_a\right)\left(\prod_{\substack{a\in J\\ a+1\notin J}} (1-\beta^2 q_a)\right). \tag{1}$$

In particular $F_0^j = 1$. The Lenart–Maeno quantization map $\mathcal{Q}$ is linear over the coefficient parameters and is specified on standard elementary monomials by

$$\mathcal{Q}\left(\prod_{j=1}^{N-1} e_{r_j}(X_1,\ldots,X_j)\right) = \prod_{j=1}^{N-1} F_{r_j}^j(X;q,\beta), \qquad 0 \leq r_j \leq j. \tag{2}$$

One takes N sufficiently large for the polynomial being quantized. The standard elementary monomials form a basis of the space with $\deg_{X_i} \le N-i$, $1 \le i < N$; this specifies the map on that space. The map is not asserted to be an algebra homomorphism. We set

$$\mathfrak{G}_u^q(x;y) = \mathcal{Q}\big(\mathfrak{G}_u(x;y)\big).$$

At $\beta = -1$ this is the usual quantum double Grothendieck normalization associated with Lenart–Maeno quantization [4, 3].

The top block appearing in our formula is

$$B_k^q(x;z) := \mathcal{Q}\left(\prod_{i=1}^{k}(x_i+z)\right) = \beta^{-k}\sum_{r=0}^{k}\big(-(1-\beta z)\big)^{k-r}F_r^k(X;q,\beta). \tag{3}$$

The apparent negative powers of β in this expression cancel. At $\beta = 0$ the block is understood by polynomial specialization, not by separately substituting into β^{-k}. We set $B_0^q = 1$. Notice that (3) quantizes the ordinary factors $x_i + z$, not the formal-group factors $x_i \oplus z$.

2 Chains and endpoint factors

Fix $u \in S_\infty$ and $k \ge 1$. Choose a finite rank $R > k$ with $u \in S_R$. The directed quantum Bruhat graph on S_R has an edge

$$v \xrightarrow{(a,b)} vt_{ab} \qquad (1 \le a < b \le R)$$

when either

$$\begin{aligned} &\ell(vt_{ab}) = \ell(v) + 1 && \text{(a Bruhat edge)},\\ &\ell(vt_{ab}) = \ell(v) - 2(b-a) + 1 && \text{(a quantum edge)}. \end{aligned}$$

A Bruhat edge has weight 1; a quantum edge labeled (a,b) has weight $q_a q_{a+1}\cdots q_{b-1}$. Order labels by

$$(a,b) \prec (c,d) \quad\Longleftrightarrow\quad b > d \ \text{ or } \ (b = d \text{ and } a < c).$$

Definition 1. *An admissible k-chain from u in S_R is a directed path*

$$\gamma: \quad u = v_0 \xrightarrow{(a_1,b_1)} v_1 \xrightarrow{(a_2,b_2)} \cdots \xrightarrow{(a_s,b_s)} v_s = w$$

in the quantum Bruhat graph satisfying:

1. *$a_i \le k < b_i$ for every i;*
2. *all labels (a_i,b_i) are distinct;*
3. *$b_1 \ge b_2 \ge \cdots \ge b_s$;*
4. *if $i < s$ and $a_i = a_j$ for some $j < i$, then $(a_i,b_i) \prec (a_{i+1},b_{i+1})$.*

The empty path, with endpoint u, is included. Let $E_R(u,k)$ be the set of endpoints of these paths. Write $s(\gamma)$ for the number of edges, and define $D(\gamma) = (D_1,\dots,D_{R-1})$ by

$$D_j(\gamma) = \#\{\, i : \text{the } i\text{th edge is quantum and } a_i \le j < b_i \,\}.$$

Thus the weight of γ is $q^{D(\gamma)} = \prod_{j=1}^{R-1} q_j^{D_j(\gamma)}$.

Directly from the edge definitions, every such path satisfies

$$s(\gamma) = \ell(w) - \ell(u) + 2\sum_{j=1}^{R-1} D_j(\gamma). \tag{4}$$

For an endpoint w define three disjoint sets of *values* from the initial window $u(1), \dots, u(k)$:

$$\begin{aligned}
\mathrm{Fix}(u, w; k) &= \{u(i) : 1 \le i \le k,\ w(i) = u(i)\},\\
\mathrm{Left}(u, w; k) &= \{u(i) : 1 \le i \le k,\ w^{-1}(u(i)) < i\},\\
\mathrm{Out}(u, w; k) &= \{u(i) : 1 \le i \le k\} \setminus \big(\mathrm{Fix}(u, w; k) \cup \mathrm{Left}(u, w; k)\big).
\end{aligned}$$

Set

$$m(u, w; k) = |\,\mathrm{Left}(u, w; k)| + |\,\mathrm{Out}(u, w; k)| = k - |\,\mathrm{Fix}(u, w; k)|.$$

The name Out includes values that move right while remaining in the window, as well as values that leave the window.

3 The triple quantum K-Pieri formula

We state the triple quantum K-Pieri formula, expanding a quantum double Grothendieck polynomial in the y-equivariant basis times a quantized top block carrying a second parameter z.

Theorem 2 (Triple quantum K-Pieri formula)**.** *Let $k \ge 1$ and $u \in S_\infty$. Set*

$$R_0 = \max\{\max(\{2\} \cup \mathrm{supp}(u)),\ k+1\} + k, \qquad E = E_{R_0}(u, k).$$

For each endpoint $w \in E$, all admissible chains from u to w have the same quantum degree $D(u, w; k)$ and length $s(u, w; k)$. Moreover, $s(u, w; k) \ge m(u, w; k)$, and

$$B_k^q(x; z)\, \mathfrak{G}_u^q(x; y) = \sum_{w \in E} C_{u,w}^{(k)}(y, z, q, \beta)\, \mathfrak{G}_w^q(x; y), \tag{5}$$

where

$$\boxed{\begin{aligned} C_{u,w}^{(k)} = q^{D(u,w;k)} \beta^{s(u,w;k) - m(u,w;k)} \prod_{a \in \mathrm{Fix}(u,w;k)} \left(z - \frac{y_a}{1 + \beta y_a}\right)\\ \times (1 - \beta z)^{|\,\mathrm{Left}(u,w;k)|} \prod_{a \in \mathrm{Out}(u,w;k)} (1 + \beta y_a)^{-1}. \end{aligned}} \tag{6}$$

Here $B_k^q(x; z) = \mathcal{Q}(\prod_{i=1}^k (x_i + z))$ is defined in (3); D records the quantum edges in a chain; s is its number of edges; and $m = |\,\mathrm{Left}\,| + |\,\mathrm{Out}\,| = k - |\,\mathrm{Fix}\,|$. The sets Fix, Left, Out *record whether each value among $u(1), \dots, u(k)$ stays fixed, moves to an earlier position, or otherwise moves (to a later position or out of the first k positions). The admissible chains and their quantum weights are defined above.*

The formula is indexed by endpoints, so each endpoint appears once. The chain-independence of D and s is part of the theorem.

Remark 3 (Rank and polynomial setting)**.** *The rank R_0 just makes the endpoint set finite; it does not impose relations from a finite flag variety. For $k = 0$ the formula reduces to $\mathfrak{G}_u^q = \mathfrak{G}_u^q$.*

Remark 4 (A useful degree check)**.** *Let* $|D| = \sum_j D_j$*. By* (4)*, the prefactor in* (6) *can also be written as*

$$q^D \beta^{s-m} = Q^D \beta^{\ell(w)-\ell(u)-m}.$$

The first expression is preferable for specialization at $\beta = 0$*, since the inequality* $s \geq m$ *makes nonnegativity of the power explicit. With* $\deg x_i = \deg y_i = \deg z = 1$*,* $\deg \beta = -1$*, and* $\deg q_i = 2$*, the coefficient has degree* $k + \ell(u) - \ell(w)$*.*

4 Proof calculations

The arguments below establish the rank bound, polynomiality of the top block, the full formula for $k = 1$, and an all-k non-equivariant marked-chain expansion. The remaining step for the general equivariant formula is specified at the end of the section; that step is not proved here.

4.1 Finiteness and the power of β

Proposition 5. *Put* $n(u) = \max(\{1\} \cup \operatorname{supp}(u))$ *and* $M = \max\{n(u), k\} + 1$*. Every label in an admissible* k*-chain from* u *has second entry at most* M*. Consequently its endpoint set stabilizes in every rank containing* u *and* $\{1, \ldots, M\}$*. For every such chain* γ *with endpoint* w*,*

$$s(\gamma) \geq m(u, w; k).$$

Proof. The first label has maximal second entry, by the ordering condition. Suppose it were (a, b) with $b > M$. Then $a \leq k < b - 1$, $u(b) = b$, $u(b-1) = b - 1$, and $u(a) < b - 1$. The intermediate value $u(b-1)$ prevents (a, b) from giving a Bruhat cover. Since $u(a) < u(b)$, the edge cannot be quantum either. This contradiction proves the rank bound. Adding fixed positions after M changes neither the relevant inversion differences nor the permitted paths, proving stabilization.

Every position $i \leq k$ with $w(i) \neq u(i)$ must occur in at least one transposition of the path. Each transposition involves exactly one position in $\{1, \ldots, k\}$. There are $k - |\operatorname{Fix}(u, w; k)| = m(u, w; k)$ changed positions, so at least this many edges are required. □

This proves that the displayed power β^{s-m} is nonnegative for each path, without presupposing independence of the path. It does not prove independence of s or D for a fixed endpoint.

4.2 An explicit polynomial top block

Write $a = 1 - \beta z$. For $S \subseteq \{1, \ldots, k\}$ with no two consecutive elements, put

$$\epsilon(S) = \begin{cases} 1 & k \in S, \\ 0 & k \notin S, \end{cases} \qquad U(S) = \{1, \ldots, k\} \setminus \big(S \cup (S+1)\big), \quad S + 1 = \{j + 1 : j \in S\}.$$

Proposition 6. *The top block has the polynomial expression*

$$B_k^q(x; z) = \sum_{\substack{S \subseteq \{1,\ldots,k\} \\ S \text{ has no consecutive elements}}} (-\beta)^{\epsilon(S)} a^{|S| - \epsilon(S)} \left(\prod_{j \in S} q_j X_j \right) \left(\prod_{i \in U(S)} (x_i + z) \right). \tag{7}$$

In particular, no inverse power of β *is needed. For* $k \geq 2$ *it satisfies*

$$\begin{aligned} (1 - \beta^2 q_{k-1}) B_k^q = \big(x_k + z + \beta(q_{k-1} - q_k) - \beta^2 (q_k x_k + z q_{k-1})\big) B_{k-1}^q \\ + (1 - \beta z) q_{k-1} (1 - \beta^2 q_k) X_k B_{k-2}^q, \end{aligned} \tag{8}$$

with $B_0^q = 1$ *and* $B_1^q = x_1 + z - \beta q_1 X_1$*.*

Proof. Multiply (3) by β^k and substitute (1). In each product of factors $1-Q_j$, choose the term $-Q_j$ for a set S of indices. Such a choice forces $j \in J$ and $j+1 \notin J$ for each $j \in S$, so two elements of S cannot be consecutive. Conversely these are the only restrictions on the remaining choices of J.

Each forced selected position contributes X_j. Each forced unselected position $j+1 \leq k$ contributes $-a$. At every other position, summing the selected and unselected possibilities gives $X_i - a$. Thus the contribution for S to $\beta^k B_k^q$ is

$$(-1)^{|S|} Q^S \left(\prod_{j \in S} X_j \right) (-a)^{|S|-\epsilon(S)} \prod_{i \in U(S)} (X_i - a), \qquad Q^S = \prod_{j \in S} Q_j.$$

Since $|U(S)| = k - 2|S| + \epsilon(S)$, $Q_j = \beta^2 q_j$, and $X_i - a = \beta(x_i + z)$, this expression is β^k times the summand in (7). This proves both that formula and polynomiality.

For the recurrence, introduce the auxiliary polynomials

$$H_0 = 1, \qquad H_1 = x_1 + z, \qquad H_j = (x_j + z) H_{j-1} + a q_{j-1} X_{j-1} H_{j-2} \quad (j \geq 2).$$

They enumerate tilings of $\{1, \ldots, j\}$ by single positions of weight $x_i + z$ and adjacent pairs $\{i, i+1\}$ of weight $a q_i X_i$. Separating the terms of (7) according to whether $k \in S$ gives

$$B_j^q = H_j - \beta q_j X_j H_{j-1} \qquad (j \geq 1). \tag{9}$$

Set $T_j = (1 - \beta^2 q_j) X_j$. The recurrence for H_j, together with $\beta(x_j + z - \beta q_j X_j) + a = T_j$, gives

$$B_k^q = (x_k + z - \beta q_k X_k) B_{k-1}^q + q_{k-1} X_{k-1} T_k H_{k-2}.$$

For $k \geq 3$, the same calculation at $k-1$ gives

$$\beta B_{k-1}^q = T_{k-1} H_{k-2} - a B_{k-2}^q.$$

For $k = 2$ this last identity follows directly from B_1^q. Multiplying the preceding expression for B_k^q by $1 - \beta^2 q_{k-1}$ and using this identity yields

$$(1 - \beta^2 q_{k-1}) B_k^q = \frac{T_k - a(1 - \beta^2 q_{k-1})}{\beta} B_{k-1}^q + a q_{k-1} T_k B_{k-2}^q.$$

Expanding the numerator gives exactly (8); it is divisible by β. □

At $\beta = 0$, this becomes

$$B_k^q = (x_k + z) B_{k-1}^q + q_{k-1} B_{k-2}^q,$$

the elementary quantum cohomological recurrence. The additional terms in (8) are therefore essential K-theoretic corrections, not changes in notation.

4.3 Passing from finite flags to polynomial identities

We will use the equivariant quantum Chevalley formula, which is initially an identity of classes on a finite flag variety. The following observation explains the passage back to polynomials.

Proposition 7 (Stable detection)**.** *At $\beta = -1$, let P be a polynomial in finitely many x_i and q_i, with coefficients rational in finitely many y_a. If its image in the localized equivariant quantum K-theory presentation is zero in every sufficiently large rank, then $P = 0$ as a polynomial.*

Proof. Suppose $P \neq 0$, and choose its least nonzero total quantum degree. In the quantum-adic expansion of its image, the coefficient of each monomial of that degree is the ordinary equivariant K-theory class of the corresponding coefficient polynomial $f(x;y)$. Indeed, all smaller quantum-degree coefficient polynomials of P are zero. The quantum K-theory module is free on the Schubert classes, so each such ordinary class must vanish.

Let $x_1, \ldots, x_r$ include the variables occurring in f, and let d bound its degree in each of these variables. Choose a rank $N > d + r$ containing all parameters in question. Restricting a vanishing ordinary class to every torus fixed point gives

$$f(\overline{y}_{\sigma(1)}, \ldots, \overline{y}_{\sigma(r)}; y) = 0 \qquad (\sigma \in S_N).$$

Work over the rational function field in the equivariant parameters. The N numbers $\overline{y}_a$ are distinct. After fixing distinct values for $r-1$ variables, the remaining variable has $N - r + 1 > d$ possible values. Polynomial interpolation says that every coefficient in that variable vanishes. Repeating this argument in the other variables proves $f = 0$, a contradiction. □

We use here the presentation and Schubert representatives of [3]. Localizing the equivariant coefficient ring does not affect the conclusion, since clearing its denominators is injective.

4.4 The line Chevalley formula

The proof of Theorem 2 is an induction on k driven by multiplication by the single line class

$$T_k = (1 - \beta^2 q_k) X_k = (1 - \beta^2 q_k)(1 + \beta x_k),$$

exactly as the proof of the quantum Schubert Pieri formula is driven by multiplication by $x_k - z$. The following lemma is the K-theoretic single-variable Monk formula.

Definition 8 (Line paths)**.** *Let $k \geq 1$ and $u \in S_\infty$, and put $M = \max\{n(u), k\} + 1$ as in Proposition 5. A* line path at k from u *is a directed path π in the quantum Bruhat graph with labels*

$$(a_1, k),\ (a_2, k),\ \ldots,\ (a_s, k), \qquad (k, b_1),\ (k, b_2),\ \ldots,\ (k, b_t),$$

where $s, t \geq 0$, $k > a_1 > a_2 > \cdots > a_s \geq 1$, and $M \geq b_1 > b_2 > \cdots > b_t > k$. Write $s(\pi)$ and $t(\pi)$ for the two lengths, $v(\pi)$ for the permutation reached after the first $s(\pi)$ edges, $\mathrm{end}(\pi)$ *for the final permutation, and $D(\pi)$ for the quantum degree, defined edgewise as in Definition 1. The empty path is allowed, with $v(\pi) = \mathrm{end}(\pi) = u$.*

Lemma 9 (Line Chevalley formula)**.** *For every $k \geq 1$ and $u \in S_\infty$,*

$$T_k\, \mathfrak{G}^q_u(x;y) = \sum_\pi (-\beta)^{s(\pi)} \beta^{t(\pi)} q^{D(\pi)} \frac{1}{1 + \beta y_{v(\pi)(k)}}\, \mathfrak{G}^q_{\mathrm{end}(\pi)}(x;y), \tag{10}$$

the sum running over all line paths at k from u.

Proof. Set $\beta = -1$ first and write $Y_a = (1 - y_a)^{-1}$, so that the claimed coefficient is $(-1)^{t(\pi)} q^{D(\pi)} Y_{v(\pi)(k)}$. Work in $QK_T(Fl_N)$ for N large, with the presentation and Schubert representatives of [3]: the residue class of T_k is the line bundle class $[\mathcal{O}(-\epsilon_k)]$ with $\epsilon_k = \varpi_k - \varpi_{k-1}$, $\mathfrak{G}^q_w$ represents the opposite Schubert class $[\mathcal{O}^w]$, and the equivariant parameters are identified by $1 - y_a = \mathbf{e}^{-\epsilon_a}$.

Kato's isomorphism Φ onto the equivariant K-group of the semi-infinite flag manifold sends $\mathbf{e}^\mu [\mathcal{O}^w] Q^\xi$ to $\mathbf{e}^{-\mu} [\mathcal{O}_{\mathbf{Q}}(wt_\xi)]$, and by [3, Section 3.1], quoting [2, Proposition 5.2], it intertwines quantum multiplication by $[\mathcal{O}(-\epsilon_k)]$ with $(1 - \mathbf{t}_{k-1})\,[\mathcal{O}_{\mathbf{Q}}(w_0 \epsilon_k)] \otimes (-)$, where $\mathbf{t}_{k-1}$ corresponds to

Q_{k-1} and $\mathbf{t}_0 = 0$. The tensor product with $[\mathcal{O}_{\mathbf{Q}}(w_0\epsilon_k)]$ is computed by the general Chevalley formula of Lenart–Naito–Sagaki [1, Theorem 33], in the form recalled in [3, Section 5.1], for the weight $\lambda = -\epsilon_k$ and any reduced λ-chain. We use the chain of [3, Section 5.2]: the alcove path from $A_\circ$ to A_{ϵ_k} crosses exactly the hyperplanes

$$H_{\epsilon_a-\epsilon_k,0}\ (a = k-1,\dots,1), \qquad H_{\epsilon_k-\epsilon_b,1}\ (b = N,\dots,k+1),$$

in this order. The first group is crossed in the negative direction, so the corresponding roots of the chain are the positive roots $\epsilon_a - \epsilon_k$; the second group is crossed in the positive direction, so the corresponding roots are the negative roots $-(\epsilon_k - \epsilon_b)$. An admissible subset is therefore a line path π, and the statistic $n(A)$ counting negative roots is $t(\pi)$. The affine reflections of the first group are linear, while those of the second group fix ϵ_k, because $\epsilon_k \in H_{\epsilon_k-\epsilon_b,1}$; hence the weight statistic is $-v(\pi)\epsilon_k = -\epsilon_{v(\pi)(k)}$. Finally, $\langle -\epsilon_k, \alpha_{k-1}^\vee\rangle = 1$ and all other pairings of $-\epsilon_k$ with simple coroots are nonpositive, so the auxiliary sum over partitions in the general formula is a geometric series contributing $(1-\mathbf{t}_{k-1})^{-1}$, which cancels the factor $1-\mathbf{t}_{k-1}$ above. Transporting back through Φ, which negates the exponent of $\mathbf{e}$ and converts t_ξ into Q^ξ, gives

$$T_k \star [\mathcal{O}^u] = \sum_\pi (-1)^{t(\pi)} Q^{D(\pi)} \mathbf{e}^{\epsilon_{v(\pi)(k)}} [\mathcal{O}^{\mathrm{end}(\pi)}] = \sum_\pi (-1)^{t(\pi)} q^{D(\pi)} Y_{v(\pi)(k)} [\mathcal{O}^{\mathrm{end}(\pi)}]$$

in $QK_T(Fl_N)$. For $k = 1$ there are no left edges and this is the first fundamental weight case of the Chevalley formula in [1, Section 5], after the substitution $[\mathcal{O}^{s_1}] = 1 - \mathbf{e}^{-\varpi_1} T_1$.

The bound $b_1 \le M$ is proved as in Proposition 5: $v(\pi)$ agrees with u outside $\{1,\dots,k\}$, so if $b_1 > M$ then $v(\pi)(b_1) = b_1$, $v(\pi)(b_1 - 1) = b_1 - 1$ and $v(\pi)(k) < b_1 - 1$, and (k, b_1) is neither a Bruhat nor a quantum edge. Thus the right side of (10) is independent of N, and Proposition 7 upgrades the identity in $QK_T(Fl_N)$ to a polynomial identity at $\beta = -1$. The homogenization to general β is the scaling used in Proposition 10 below: with $c = -\beta$ and $(x, y, q) \mapsto (cx, cy, \beta^2 q)$ one has $T_k \mapsto T_k$, $Y_a \mapsto (1+\beta y_a)^{-1}$, $\mathfrak{G}_w^q \mapsto c^{\ell(w)} \mathfrak{G}_w^q$, and

$$c^{\ell(\mathrm{end}(\pi))-\ell(u)} \beta^{2|D(\pi)|} (-1)^{t(\pi)} = (-\beta)^{s(\pi)} \beta^{t(\pi)}$$

by the length–degree identity $\ell(\mathrm{end}(\pi)) - \ell(u) = s(\pi) + t(\pi) - 2|D(\pi)|$. □

We record the operator form in which the lemma is used. For a formal combination $\Theta = \sum_v \theta_v \mathfrak{G}_v^q$ whose coefficients are free of x, put

$$\mathcal{T}_k \Theta := \sum_v \theta_v \sum_{\pi \text{ from } v} (-\beta)^{s(\pi)} \beta^{t(\pi)} q^{D(\pi)} \frac{\mathfrak{G}^q_{\mathrm{end}(\pi)}}{1 + \beta y_{v(\pi)(k)}}, \tag{11}$$

so that Lemma 9 reads $T_k\Theta = \mathcal{T}_k\Theta$ as polynomials.

4.5 Proof of the triple formula for $k = 1$

Proposition 10. *Theorem 2 holds for $k = 1$ and arbitrary $u \in S_\infty$.*

Proof. Take $k = 1$ in Lemma 9. There are no left edges, so $v(\pi) = u$ for every line path, the equivariant factor is the constant $(1+\beta y_{u(1)})^{-1}$, and

$$(1-\beta^2 q_1) X_1 \mathfrak{G}_u^q = \frac{1}{1+\beta y_{u(1)}} \sum_\gamma \beta^{s(\gamma)} q^{D(\gamma)} \mathfrak{G}^q_{\mathrm{end}(\gamma)}, \tag{12}$$

where γ runs over directed paths with labels forming a subsequence of $(1,M),(1,M-1),\dots,(1,2)$. These are exactly the admissible chains of Definition 1 for $k=1$: all lower indices are 1, and distinctness and weak decrease of the upper indices make them strictly decreasing. There is at most one such path to a given endpoint. In fact, a nonempty product $t_{1b_1}\cdots t_{1b_s}$ with distinct b_i is a single cycle on $\{1,b_1,\dots,b_s\}$. The cycle of $u^{-1}w$ determines the ordered list of these indices. It also shows that a nonempty path cannot have endpoint u, and that the value $u(1)$ has moved out of position 1. Consequently a nonempty path has $m=1$, $\mathrm{Fix}=\mathrm{Left}=\varnothing$, and $\mathrm{Out}=\{u(1)\}$.

Finally, $\beta B_1^q=(1-\beta^2q_1)X_1-(1-\beta z)$. Subtract the second term from (12). The empty-path coefficient, after division by β, is

$$\frac{(1+\beta y_{u(1)})^{-1}-(1-\beta z)}{\beta}=z-\frac{y_{u(1)}}{1+\beta y_{u(1)}}.$$

Every nonempty path has coefficient

$$q^{D(\gamma)}\frac{\beta^{s(\gamma)-1}}{1+\beta y_{u(1)}}.$$

These are precisely (6) in the two cases. All exponents are nonnegative, so clearing equivariant denominators extends the result to $\beta=0$ as well. □

For later use we make the scaling in the proof of Lemma 9 explicit. Put $c=-\beta$ and replace (x,y,z,q) at $\beta=-1$ by (cx,cy,cz,β^2q). The defining divided differences and the quantization map give

$$\mathfrak{G}_w^{\beta^2q}(cx;cy)\big|_{\beta=-1}=c^{\ell(w)}\mathfrak{G}_w^q(x;y), \tag{13}$$

where on the left the quantum arguments are β^2q and the Grothendieck parameter is fixed at -1. The scaling takes place with β inverted; whenever both sides of a resulting identity have no negative powers of β, it holds before that inversion as well.

4.6 What the published marked-chain formula proves for all k

There is also an unconditional all-k calculation, but it is non-equivariant and initially indexed by paths rather than endpoints. For a nonempty admissible chain γ, let $r(\gamma)$ be the number of distinct lower indices a_i. Let $F(\gamma)$ be the following set of edge indices:

- every i in the maximal initial segment with $b_1=\cdots=b_i$ and $a_1>\cdots>a_i$;
- every $i<s(\gamma)$ for which $(a_i,b_i)\succ(a_{i+1},b_{i+1})$.

Put $h(\gamma)=|F(\gamma)|$. These are exactly the indices forced to be marked in the Naito–Sagaki marking rule; in particular $h(\gamma)\ge 1$.

Proposition 11. *For every $k\ge 1$ and $u\in S_\infty$,*

$$B_k^q(x;z)\mathfrak{G}_u^q(x;0)=z^k\mathfrak{G}_u^q(x;0)+\sum_{\substack{\gamma\ \mathit{admissible}\\ s(\gamma)>0}}q^{D(\gamma)}\beta^{s(\gamma)-r(\gamma)}z^{k-r(\gamma)}(1-\beta z)^{h(\gamma)-1}\mathfrak{G}^q_{\mathrm{end}(\gamma)}(x;0). \tag{14}$$

This statement does not require uniqueness of a path to its endpoint.

Proof. For $1 \le p \le k$, let $c[k,p] = (k-p+1, k-p+2, \dots, k+1)$, in cycle notation. The single Grothendieck polynomial of this cycle is

$$\mathfrak{G}_{c[k,p]}(x;0) = \sum_{j=p}^{k} \binom{j-1}{p-1} \beta^{j-p} e_j(x_1, \dots, x_k).$$

The binomial theorem, coefficient by coefficient in e_j, gives

$$\prod_{i=1}^{k}(x_i + z) = z^k + \sum_{p=1}^{k} z^{k-p}(1-\beta z)^{p-1}\mathfrak{G}_{c[k,p]}(x;0).$$

Indeed the coefficient of e_j on the right is $z^{k-j}\big((1-\beta z)+\beta z\big)^{j-1} = z^{k-j}$. Applying the *linear* map $\mathcal{Q}$ gives

$$B_k^q = z^k + \sum_{p=1}^{k} z^{k-p}(1-\beta z)^{p-1}\mathfrak{G}^q_{c[k,p]}(x;0). \tag{15}$$

No multiplicativity of $\mathcal{Q}$ is used.

Naito–Sagaki's Pieri theorem [4, Theorem 2.13], homogenized from $\beta = -1$, says that the contribution of γ to $\mathfrak{G}^q_u \mathfrak{G}^q_{c[k,p]}$ is

$$q^{D(\gamma)}\beta^{s(\gamma)-p} \# \operatorname{Mark}_p(\gamma).$$

A mark may occur only at a first occurrence of a lower index. All indices in $F(\gamma)$ must be marked; the remaining first occurrences can be chosen freely. The admissibility conditions ensure that forced indices are first occurrences. Thus, as also recorded in [4, Remark 2.14],

$$\# \operatorname{Mark}_p(\gamma) = \binom{r(\gamma)-h(\gamma)}{p-h(\gamma)}.$$

Here $1 \le h(\gamma) \le r(\gamma) \le k$.

Fix a nonempty chain, and abbreviate its statistics by s, r, h, D. Its coefficient after multiplying (15) by $\mathfrak{G}^q_u$ is therefore

$$\begin{aligned} & q^D \sum_{p=h}^{r} \binom{r-h}{p-h} \beta^{s-p} z^{k-p}(1-\beta z)^{p-1} \\ &\quad = q^D \beta^{s-r} z^{k-r}(1-\beta z)^{h-1} \sum_{p=h}^{r} \binom{r-h}{p-h} (\beta z)^{r-p}(1-\beta z)^{p-h} \\ &\quad = q^D \beta^{s-r} z^{k-r}(1-\beta z)^{h-1}. \end{aligned}$$

The last sum is 1 by the binomial theorem. The separate term $z^k \mathfrak{G}^q_u$ supplies the empty path. This proves (14). Homogenization is justified by the same scaling and length–degree identity used in Proposition 10; the exponents $s-p$ are nonnegative. □

4.7 Reduction of level k to levels $k-1$ and $k-2$

We now set up the induction on k. Throughout, $a = 1-\beta z$ and H_j denotes the tiling polynomial introduced in the proof of Proposition 6:

$$H_0 = 1, \qquad H_1 = x_1 + z, \qquad H_j = (x_j + z)H_{j-1} + a\, q_{j-1} X_{j-1} H_{j-2} \quad (j \ge 2),$$

and $B^q_j = H_j - \beta q_j X_j H_{j-1}$ for $j \ge 1$ by (9). We use the convention $q_0 = 0$ and $B^q_{-1} = 0$.

Lemma 12 (Two-step block identities)**.** *For every $k \geq 1$,*

$$\beta B^q_k + a B^q_{k-1} = T_k H_{k-1}, \tag{16}$$

$$(1-\beta^2 q_{k-1}) H_{k-1} = B^q_{k-1} + a\beta q_{k-1} B^q_{k-2}. \tag{17}$$

Consequently

$$(1-\beta^2 q_{k-1})\big(\beta B^q_k + a B^q_{k-1}\big) = T_k\big(B^q_{k-1} + a\beta q_{k-1} B^q_{k-2}\big). \tag{18}$$

Proof. Substituting $B^q_k = H_k - \beta q_k X_k H_{k-1}$, $B^q_{k-1} = H_{k-1} - \beta q_{k-1} X_{k-1} H_{k-2}$ and the recurrence for H_k,

$$\beta B^q_k + a B^q_{k-1} = \big(\beta(x_k+z) + a - \beta^2 q_k X_k\big) H_{k-1} = (1-\beta^2 q_k) X_k H_{k-1},$$

since $\beta(x_k+z)+a = 1+\beta x_k = X_k$; the two terms involving $a\beta q_{k-1} X_{k-1} H_{k-2}$ cancel. This is (16). For (17) with $k \geq 3$, expand both sides using the recurrence for H_{k-1} and the expressions of B^q_{k-1}, B^q_{k-2} through H: the right side is

$$H_{k-1} + \beta q_{k-1}(a - X_{k-1}) H_{k-2} - a\beta^2 q_{k-1} q_{k-2} X_{k-2} H_{k-3},$$

and $\beta(a - X_{k-1}) = -\beta^2(x_{k-1}+z)$, so this equals $H_{k-1} - \beta^2 q_{k-1}\big((x_{k-1}+z) H_{k-2} + a q_{k-2} X_{k-2} H_{k-3}\big) = (1-\beta^2 q_{k-1}) H_{k-1}$. For $k=2$ the same computation applies without the last term, and for $k=1$ both sides equal 1. Multiplying (16) by $1-\beta^2 q_{k-1}$ and substituting (17) gives (18). □

At $\beta = 0$, (18) reduces to $B_k = (x_k+z) B_{k-1} + q_{k-1} B_{k-2}$, the recurrence used for the quantum Schubert Pieri formula. At $\beta = -1$ it reads $(1-q_{k-1})\big(A B_{k-1} - B_k\big) = T_k\big(B_{k-1} - A q_{k-1} B_{k-2}\big)$ with $A = 1+z$.

The target identity. For $j \geq 0$ and $u \in S_\infty$ write

$$P_j(u) := \sum_{w \in E(u,j)} C^{(j)}_{u,w}\, \mathfrak{G}^q_w, \qquad P_0(u) = \mathfrak{G}^q_u, \qquad P_{-1}(u) = 0,$$

for the right side of (5) at level j, with $C^{(j)}_{u,w}$ given by (6). Here the endpoint data (s, D) entering $C^{(j)}_{u,w}$ are assumed well defined; this is part of what must be proved at each level, see Remark 14 below.

Proposition 13 (Reduction)**.** *Let $k \geq 1$ and suppose Theorem 2 holds at levels $k-1$ and $k-2$ for the permutation u, that is, $B^q_{k-1}\mathfrak{G}^q_u = P_{k-1}(u)$ and $B^q_{k-2}\mathfrak{G}^q_u = P_{k-2}(u)$. Then Theorem 2 holds at level k for u if and only if*

$$\beta(1-\beta^2 q_{k-1})\, P_k(u) = \mathcal{T}_k\big(P_{k-1}(u) + a\beta q_{k-1} P_{k-2}(u)\big) - a(1-\beta^2 q_{k-1})\, P_{k-1}(u), \tag{19}$$

with $\mathcal{T}_k$ the line operator (11)*.*

Proof. By (18) and the hypotheses,

$$(1-\beta^2 q_{k-1})\big(\beta B^q_k \mathfrak{G}^q_u + a P_{k-1}(u)\big) = T_k\big(P_{k-1}(u) + a\beta q_{k-1} P_{k-2}(u)\big),$$

and the right side equals $\mathcal{T}_k\big(P_{k-1}(u) + a\beta q_{k-1} P_{k-2}(u)\big)$ by Lemma 9. Hence $\beta(1-\beta^2 q_{k-1}) B^q_k \mathfrak{G}^q_u$ equals the right side of (19). Since the $\mathfrak{G}^q_w$ are linearly independent over the coefficient field and $\beta(1-\beta^2 q_{k-1})$ is a nonzero element of the integral domain of coefficients, $B^q_k \mathfrak{G}^q_u = P_k(u)$ holds if and only if (19) does. □

For $k=1$ the identity (19) is $\beta P_1(u) = \mathcal{T}_1 \mathfrak{G}^q_u - a\mathfrak{G}^q_u$, which is exactly the computation in the proof of Proposition 10. Thus Theorem 2 follows, by induction on k, from (19) for all $k \geq 2$ and all u. We call (19) the *junction identity*; it is the K-theoretic counterpart of the recurrence $\mathcal{E}_{k,k} = (x_k - z_1)\mathcal{E}_{k-1,k-1} + q_{k-1}\mathcal{E}_{k-2,k-2}$ for quantum factorial elementary polynomials expanded by the single-variable Monk formula in the quantum Schubert case.

Coefficient form. Extracting the coefficient of $\mathfrak{G}_w^q$ in (19), and writing $\pi\colon v \to w$ for a line path at k from v with endpoint w, the junction identity reads

$$\begin{aligned}\beta(1-\beta^2 q_{k-1})\, C_{u,w}^{(k)} = \sum_{v} \sum_{\pi\colon v\to w} \frac{(-\beta)^{s(\pi)}\beta^{t(\pi)} q^{D(\pi)}}{1+\beta y_{v(\pi)(k)}} \Big(C_{u,v}^{(k-1)} + a\beta q_{k-1} C_{u,v}^{(k-2)} \Big) \\ - a(1-\beta^2 q_{k-1})\, C_{u,w}^{(k-1)},\end{aligned} \tag{20}$$

where $C_{u,v}^{(j)} = 0$ for $v \notin E(u,j)$ and $C^{(-1)} = 0$. A line path $\pi\colon v \to w$ is determined by its label sequence, and $v = w\, t_{kb_t} \cdots t_{kb_1} t_{a_s k} \cdots t_{a_1 k}$; the path must be directed from v in the quantum Bruhat graph. The term $\pi = \varnothing$, $v = w$, is the diagonal term, with equivariant factor $(1+\beta y_{w(k)})^{-1}$; the paths with $s(\pi) = 0$ are the up-neighbours $v = w t_{kb_t} \cdots t_{kb_1}$, and the paths with $t(\pi) = 0$ are the down-neighbours $v = w t_{a_s k} \cdots t_{a_1 k}$. In the quantum Schubert case only single edges occur and (20) becomes the four-term expansion of that proof; here a path may combine several down edges followed by several up edges, and each term carries a K-theoretic sign and an equivariant denominator determined by the intermediate permutation $v(\pi)$.

Remark 14 (Well-definedness)**.** *The identity* (20) *is to be proved for each endpoint w. If some w were reached by two admissible k-chains with different (s, D), the left side would be ambiguous. Hence a proof of* (20) *must also show that, for each w, the right side is a single monomial in q times the expected endpoint factors; this establishes the well-definedness claimed in Theorem 2 at level k along the way. The same situation occurs in the quantum Schubert proof, where path-independence of the quantum weight is proved through the cycle structure of $u^{-1}w$.*

The diagonal case. Suppose $u(k) = w(k)$. Position k is touched by a k-chain only through labels (k, b), and once touched it never regains its original value: the value $u(k)$ moves to a position $b > k$ that is never touched again, because later labels have second entry at most b and distinct labels cannot repeat (k, b). Hence the admissible k-chains from u to w are exactly the admissible $(k-1)$-chains from u to w all of whose labels have second entry at least $k+1$, and for these chains the conditions of Definition 1 at levels k and $k-1$ coincide. The same argument at level $k-1$ shows that the $(k-1)$-chains to w with some label (a, k) have $w(k) \neq u(k)$. Therefore

$$\mathrm{Fix}(u, w; k) = \mathrm{Fix}(u, w; k-1) \sqcup \{u(k)\}, \qquad \text{Left, Out, } s, \ D \text{ unchanged},$$

and

$$C_{u,w}^{(k)} = \Big(z - \frac{y_{u(k)}}{1+\beta y_{u(k)}} \Big) C_{u,w}^{(k-1)} = \frac{(1+\beta y_{u(k)})^{-1} - a}{\beta}\, C_{u,w}^{(k-1)}. \tag{21}$$

Inserting (21) into (20), the diagonal term $\pi = \varnothing$ together with the last term produce exactly $\beta C_{u,w}^{(k)}$ plus $\frac{a\beta q_{k-1}}{1+\beta y_{u(k)}} C_{u,w}^{(k-2)}$, and what remains to be shown in the case $u(k) = w(k)$ is the cancellation

$$\sum_{v} \sum_{\substack{\pi\colon v\to w \\ \pi\neq\varnothing}} \frac{(-\beta)^{s(\pi)}\beta^{t(\pi)} q^{D(\pi)}}{1+\beta y_{v(\pi)(k)}} \Big(C_{u,v}^{(k-1)} + a\beta q_{k-1} C_{u,v}^{(k-2)} \Big) + \frac{\beta q_{k-1}}{1+\beta y_{u(k)}} \Big(\beta C_{u,w}^{(k-1)} + a C_{u,w}^{(k-2)} \Big) = 0, \tag{22}$$

where we used $\beta(z + \overline{y}_{u(k)}) + a = (1+\beta y_{u(k)})^{-1}$ to combine the two multiples of $q_{k-1} C_{u,w}^{(k-1)}$. This is the K-theoretic form of the reflection–drop cancellation of the quantum Schubert proof: there the nonempty paths are single edges t_{ak}, t_{kb} and the drop term is $q_{k-1} c_{k-2}(u, w)$. Note the combination $\beta C_{u,w}^{(k-1)} + a C_{u,w}^{(k-2)}$, the coefficient analogue of $\beta B_{k-1}^q + a B_{k-2}^q = T_{k-1} H_{k-2}$.

4.8 The cancellation mechanism: cut composites and local moves

Every term of (20) is a *cut composite*: a directed path in the quantum Bruhat graph

$$u = v_0 \to v_1 \to \cdots \to v_r = v \;\big|\; v \to \cdots \to w,$$

whose first part is an admissible chain at level $k-1$ or $k-2$ and whose second part is a line path at k; the bar marks the cut. The term attached to it is the product of the chain coefficient $C^{(j)}_{u,v}$ (times $a\beta q_{k-1}$ when $j = k-2$) and the line weight $(-\beta)^s \beta^t q^D (1+\beta y_{v(\pi)(k)})^{-1}$. The whole edge sequence from u to w determines q^D for the chain and the path together, and it determines $v(\pi)$; only the position of the cut and the level distinguish the bookkeeping. The cancellation in (20) is realized by two moves on cut composites, which we now make precise, together with the identity $\beta(z+\overline{y}_c)+a = (1+\beta y_c)^{-1}$ used in (21).

Lemma 15 (Cut sliding). *Let γ be an admissible $(k-1)$-chain from u to v, let $\pi = (k-1,k)\cdot\pi'$ be a line path at k from v whose first edge is $(k-1,k)$, and put $v' = vt_{k-1,k}$ and $\kappa \in \{1, q_{k-1}\}$ for the weight of the edge $v \to v'$. Then:*

1. *$\gamma' = \gamma\cdot(k-1,k)$ is an admissible $(k-1)$-chain from u to v' if and only if $(k-1,k)\notin\gamma$, and π' is a line path at k from v' with $v(\pi') = v(\pi)$, $s(\pi') = s(\pi)-1$, $t(\pi') = t(\pi)$ and $q^{D(\pi)} = \kappa\, q^{D(\pi')}$.*
2. *If $v(k-1) \neq u(k-1)$ and $v(k) \neq u(k-1)$, then* Fix, Left, Out *computed for v and for v' agree, so that $C^{(k-1)}_{u,v'} = \beta\kappa\, C^{(k-1)}_{u,v}$, and the two terms of (20) indexed by (v,π) and (v',π') cancel exactly.*
3. *If $v(k-1) = u(k-1)$, then γ is also an admissible $(k-2)$-chain to v, $C^{(k-1)}_{u,v} = (z+\overline{y}_{u(k-1)})C^{(k-2)}_{u,v}$, and the two terms indexed by (v,π) and (v',π') sum to*

$$a\kappa\,(-\beta)^{s(\pi')}\beta^{t(\pi')}q^{D(\pi')}\,\frac{C^{(k-2)}_{u,v}}{1+\beta y_{v(\pi)(k)}}. \tag{23}$$

Proof. (1) The edge $(k-1,k)$ joins adjacent positions, so it is always an edge of the quantum Bruhat graph, Bruhat if $v(k-1) < v(k)$ and quantum with weight q_{k-1} otherwise; it is the same edge in γ' and in π, whence the statement about weights. In γ' the second entries remain weakly decreasing since k is the least possible value, and the only condition of Definition 1 that involves the new last label is distinctness; condition 4 for the previous label (a_r, b_r) is automatic, because $b_r > k$, or $b_r = k$ and $a_r \neq k-1$. Removing the first edge of a line path leaves a line path with the same left-part endpoint $v(\pi)$.

(2) Passing from v to v' exchanges the values $c = v(k-1)$ and $d = v(k)$. The value c leaves the window $\{1,\dots,k-1\}$: it cannot lie in Left (a value in position $k-1$ cannot have moved left), and if it lies in Out or is not a window value of u at all, its status is unchanged. The value d enters position $k-1$: if $d = u(i)$ with $i \le k-2$ it was in Out and remains there, and otherwise it is not a window value. Hence m is unchanged while the chain length grows by one and the quantum weight is multiplied by κ, giving $C^{(k-1)}_{u,v'} = \beta\kappa C^{(k-1)}_{u,v}$. The line weights of π and π' differ by the factor $(-\beta)\kappa$, so the two terms are negatives of each other.

(3) Here position $k-1$ is untouched by γ, so no label of γ has first entry $k-1$, and γ is admissible at level $k-2$ with the same data; this gives the factorization of $C^{(k-1)}_{u,v}$ as in (21). In v' the value $c = u(k-1)$ has moved to position k, so it passes from Fix to Out while d behaves as in (2); thus

$C^{(k-1)}_{u,v'} = \kappa\,(1+\beta y_c)^{-1}(z+\overline{y}_c)^{-1}C^{(k-1)}_{u,v}$. Writing $X = (-\beta)^{s(\pi')}\beta^{t(\pi')}q^{D(\pi')}(1+\beta y_{v(\pi)(k)})^{-1}$, the term of (v,π) is $X(-\beta)\kappa C^{(k-1)}_{u,v}$ and the term of (v',π') is $X\kappa(1+\beta y_c)^{-1}(z+\overline{y}_c)^{-1}C^{(k-1)}_{u,v}$. Their sum is

$$X\kappa\,\frac{(1+\beta y_c)^{-1}-\beta(z+\overline{y}_c)}{z+\overline{y}_c}\,C^{(k-1)}_{u,v} = X\kappa a\,C^{(k-2)}_{u,v},$$

which is (23). □

The case $v(k) = u(k-1)$, in which a fixed window value would enter position $k-1$, does not occur in a cut composite whose endpoint satisfies $w(k) = u(k)$: it did not appear in any of the verified ranges below, and we do not use it. The residue (23) has exactly the shape of a level-$(k-2)$ term of (20), up to the factor βq_{k-1}: the missing q_{k-1} appears either through $\kappa = q_{k-1}$ or through a factor q_{k-1} inside $q^{D(\pi')}$, and the missing β through the line path π' being one edge shorter. This is why the junction input is $P_{k-1}(u) + a\beta q_{k-1}P_{k-2}(u)$ and not $P_{k-1}(u)$ alone.

Lemma 16 (Braid move)**.** *Let $a < k-1$. For any permutation v, the two-edge sequences $(k-1,k),(a,k)$ and $(a,k-1),(k-1,k)$ from v have the same endpoint $vt_{k-1,k}t_{ak} = vt_{a,k-1}t_{k-1,k}$, and if both are directed paths in the quantum Bruhat graph they have the same quantum weight. Consequently, if γ is an admissible $(k-2)$-chain to v with $\gamma\cdot(a,k-1)$ admissible, and $\pi = (k-1,k)\cdot(a,k)\cdot\pi''$ is a line path from v, then $(k-1,k)\cdot\pi''$ is a line path from $vt_{a,k-1}$ with the same $v(\pi)$, and the terms indexed by (v,π) and $(vt_{a,k-1},(k-1,k)\cdot\pi'')$ at level $k-2$ cancel exactly whenever the window statistics of v and $vt_{a,k-1}$ agree.*

Proof. The identity $t_{k-1,k}t_{ak} = t_{a,k-1}t_{k-1,k}$ holds in S_∞. Both sequences are paths of length two between the same vertices, and $v^{-1}vt_{k-1,k}t_{ak}$ is a 3-cycle, so neither endpoint is reachable in fewer than two steps; by Postnikov's lemma all shortest paths between two vertices of the quantum Bruhat graph have the same weight. The label $(a,k-1)$ is admissible at level $k-2$ since $a \le k-2 < k-1$, and second entries remain weakly decreasing because $k-1$ is the least possible value. The chain gains one edge (a factor β when m is unchanged) while the line path loses one left edge (a factor $-\beta$), and the quantum weights agree, so the two terms cancel. □

When the window statistics change in Lemma 16 (the value $v(a)$ or $v(k-1)$ is a fixed window value), a residue of the same kind as (23) is produced instead.

The drop term. The diagonal term of (20), $\pi = \varnothing$ and $v = w$, contributes $(1+\beta y_{w(k)})^{-1}\big(C^{(k-1)}_{u,w} + a\beta q_{k-1}C^{(k-2)}_{u,w}\big)$, and after (21) what is left of it is the drop term in (22). In the language of cut composites it is the composite with an empty line path, and Lemma 15 pairs it with the composite $u \xrightarrow{\gamma_w} w \xrightarrow{(k-1,k)} wt_{k-1,k} \;\big|\; (k-1,k)$, for which the move of (1) is blocked because the label $(k-1,k)$ would be repeated. Its term is $-\beta\kappa'\,(1+\beta y_{u(k)})^{-1}C^{(k-1)}_{u,wt_{k-1,k}}$ with $\kappa\kappa' = q_{k-1}$, and by the computation in the proof of Lemma 15 it equals $-\beta^2 q_{k-1}(1+\beta y_{u(k)})^{-1}C^{(k-1)}_{u,w}$ when $w(k-1) \ne u(k-1)$, and $-\beta q_{k-1}(1+\beta y_{u(k)})^{-1}(1+\beta y_{u(k-1)})^{-1}C^{(k-2)}_{u,w} = -\beta q_{k-1}(1+\beta y_{u(k)})^{-1}\big(\beta C^{(k-1)}_{u,w} + aC^{(k-2)}_{u,w}\big)$ when $w(k-1) = u(k-1)$, using (21) at level $k-1$. In both cases it is the negative of the drop term, provided $C^{(k-2)}_{u,w} = 0$ in the first case.

A worked class. Take $u = 21345$, $k = 3$ and the endpoint $w = 12345$, which is not in $E(u,3)$, so the right side of (20) must vanish. The nonzero terms are indexed by three cut composites of the

single edge sequence $(1,3),(2,3),(1,3)$ from u to w at level 2, its braid image at level 1, and the drop:

$$\begin{array}{lll}
\text{level 2}: & (1,3)\,\big|\,(2,3),(1,3) & \beta^2 q_1 q_2\,(z+\overline{y}_1)\,t_2^{-1}t_3^{-1},\\
\text{level 2}: & (1,3),(2,3)\,\big|\,(1,3) & -\beta q_1 q_2\,t_1^{-1}t_2^{-1}t_3^{-1},\\
\text{level 1}: & (1,3)\,\big|\,(2,3),(1,3) & a\beta^3 q_1 q_2^2\,t_2^{-1}t_3^{-1},\\
\text{level 1}: & (1,3),(1,2)\,\big|\,(2,3) & -a\beta^3 q_1 q_2^2\,t_2^{-1}t_3^{-1},\\
\text{drop}: & & a\beta q_1 q_2\,t_2^{-1}t_3^{-1},
\end{array}$$

with $t_c = 1+\beta y_c$. The two level-1 terms are a braid pair (Lemma 16) and cancel. The two level-2 terms are a cut sliding pair of type (3) in Lemma 15, since the slid value $v(2) = 1 = u(2)$ is fixed; their sum is the residue $-a\beta q_1 q_2 t_2^{-1}t_3^{-1}$, which cancels the drop term. The residue is absorbed by the drop rather than by a level-1 path term because the relevant level-1 endpoint is w itself.

Status. Lemmas 15 and 16 are proved. What remains for the diagonal case is the global statement: the cut composites with a given endpoint w, $w(k) = u(k)$, decompose into classes generated by the two moves, and within each class the exact cancellations and the residues of types (23) sum to zero against the level-$(k-2)$ terms and the drop. We have verified this termwise, with the symbolic coefficients, for all $u \in S_5$ and all k, and for a sample of $u \in S_6$ with $k \le 5$; the identity (22) holds for every endpoint w in these ranges, including endpoints outside $E(u,k)$, where it asserts the vanishing of the right side. A proof of the global statement, and of its analogue for the moved cases below, would complete the induction.

The moved cases. Suppose $u(k) \neq w(k)$. Then $C_{u,w}^{(k-1)}$ and $C_{u,w}^{(k-2)}$ need not vanish, in contrast to the quantum Schubert case: a $(k-1)$-chain may move position k through a label (a,k). The identity (20) must then be established by matching each k-chain to w with a $(k-1)$- or $(k-2)$-chain to some v followed by a line path $v \to w$, and by cancelling all remaining terms in pairs. The $k=1$ case shows the pattern: there $P_{k-2} = 0$, every nonempty line path is a single up-path, and the matching is the identity. The complete case analysis for $k \ge 2$ is not carried out in this version.

5 A small example and specializations

Example 17. *Take $u = 21$ and $k = 1$. The endpoints are $21, 312, 132$, and the identity permutation e. Their chain data (s, D) are respectively*

$$(0,0), \qquad (1,0), \qquad (2,(1,0,\ldots)), \qquad (1,(1,0,\ldots)).$$

For instance, the path to 132 is

$$213 \xrightarrow{(1,3)} 312 \xrightarrow{(1,2)} 132,$$

whose second edge is quantum. Since $B_1^q = x_1 + z - \beta q_1(1+\beta x_1)$, the formula reads

$$\begin{aligned}
\big(x_1 + z - \beta q_1(1+\beta x_1)\big)\mathfrak{G}_{21}^q = &\left(z - \frac{y_2}{1+\beta y_2}\right)\mathfrak{G}_{21}^q\\
&+ \frac{\mathfrak{G}_{312}^q + \beta q_1 \mathfrak{G}_{132}^q + q_1}{1+\beta y_2}.
\end{aligned} \tag{24}$$

Here $\mathfrak{G}_e^q = 1$, and all polynomials have alphabets $(x;y)$. This particular identity can be verified by expanding the four quantum double Grothendieck polynomials and clearing the denominator.

Classical specialization. When all $q_i = 0$, only paths without quantum edges contribute. The multiplier becomes $\prod_{i=1}^{k}(x_i + z)$, giving the classical double Grothendieck top-block formula as a specialization of Theorem 2. In particular, the equivariant classical specialization is itself new, rather than an appeal to a previously published top-block formula.

Non-equivariant specialization. When all $y_a = 0$, the coefficient becomes

$$q^D \beta^{s-m} z^{|\,\mathrm{Fix}\,|}(1-\beta z)^{|\,\mathrm{Left}\,|}.$$

In addition setting $z = 0$ leaves only endpoints with $\mathrm{Fix} = \varnothing$, and the surviving coefficient is $q^D\beta^{s-k}$. At $\beta = -1$ it is $(-1)^{s-k}q^D$. This is the specialization to compare with the quantum Grothendieck Pieri formula of Naito–Sagaki [4]; identifying the indexing sets precisely is part of that comparison.

Cohomological specialization. At $\beta = 0$, terms with $s > m$ vanish. Those with $s = m$ have coefficient

$$q^D \prod_{a \in \mathrm{Fix}} (z - y_a).$$

The corresponding comparison is with the top quantum double Schubert block in coefficient alphabet $-y$.

Usual K-theory variables. At $\beta = -1$ the multiplicative equivariant variables in these conventions may be written

$$Y_a = \frac{1}{1 - y_a}.$$

Thus comparisons with formulas in the Y variables use $y_a = 1 - Y_a^{-1}$. This change of variables does not alter the permutation indices or the quantum parameters q_i.

References

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